

relation between respiratory diseases and air pollution

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Set-up your environment

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.1      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.3      v tibble    3.3.1
## v lubridate  1.9.5      v tidyr     1.3.2
## v purrr      1.2.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(yaml)
library(rmarkdown)
library(janitor)
```

```
##
## Attaching package: 'janitor'
##
## The following objects are masked from 'package:stats':
##
##      chisq.test, fisher.test
```

```
library(naniar)
library(tinytex)
library(ggplot2)
```

```
require(tidyverse)
require(rmarkdown)
require(yaml)
library(tidyr)
library(writexl)
library(readxl)
library(sf)
```

```
## Linking to GEOS 3.14.1, GDAL 3.12.1, PROJ 9.7.1; sf_use_s2() is TRUE
```

```
library(rnaturalearth)  
library(rnaturalearthdata)
```

```
##  
## Attaching package: 'rnaturalearthdata'  
##  
## The following object is masked from 'package:rnaturalearth':  
##  
##      countries110
```

```
library(ggmap)  
library(countrycode)  
library(dplyr)
```

Title Page

The relation between respiratory diseases

Group 2.2

Fatima Amankour # Part 1 - Identify a Social problem

1.1 Describe the Social Problem

Air pollution is the biggest current environmental health risk in the European Union, according to the European Environment Agency. The main risk of air pollution is respiratory disease, diseases like asthma or COPD. Each year, over 8 million people die from respiratory diseases (UT Health-San Antonio and South Texas Veterans Health Care System, 2021). In this paper, the relation between the severity of the air pollution and the frequency of respiratory disease is researched.

Part 2 - Data Sourcing

2.1 Load in the data

```
#loading the database  
ghs <- read.csv("ghs.csv")  
gaq <- read.csv("gaq.csv")
```

2.2 Provide a short summary of the dataset(s)

```
#head(dataset)
```

We have picked the information From 2 datasets the ghs(general health statistics) and gaq(general air quality). GHS is a datasets that is made by a Student at National Engineering College in India and gaq

is made by a Student at N.S.T from India. Both used data from credible source like the World Health Organization. Gaq includes information about air pollution from 24 countries. Ghs includes information about different types of air pollution from 20 countries.

```
inline_code = TRUE
```

2.3 Describe the type of variables included

The type of variables in ghs are countries, year, disease name, disease category, prevalence rate, incidence rate, mortality rate, age group, gender, population affected, recovery rate and others about the health care in the countries. The data types are mostly double floating point numbers and the ones where there is no zero value are factors.

The type of variables in gaq are country, year, dates, aiq bucket, all the pollution values, deforestation rates. The data types are mostly double floating point numbers and the ones where there is no zero value are factors.

For the sake of this example, I will continue with the assignment...

Part 3 - Quantifying

3.1 Data cleaning

For the data cleaning, we made sure that all the names are consistent and correct. used the correct datatypes per column of both datasets. For the ghs dataset, we deleted the other disease category's than respiratory. deleted years and countries that are not in both datasets. We deleted states and cities in the gaq dataset. We deleted columns that are not useful in the datasets. Brought all types of respiratory diseases in a year together. made the names of countries consistent in both datasets. At last, we selected all columns that we wanted to merge together and merged them.

```
# making sure every name is consistent and correct.
ghs <- janitor::clean_names(ghs)
gaq <- janitor::clean_names(gaq)

#Using the correct data types per column for ghs and gaq.
ghs <-
  ghs |>
  mutate(country = as_factor(country),
         gender = as_factor(gender),
         disease_name = as_factor(disease_name),
         disease_category = as_factor(disease_category),
         age_group = as_factor(age_group),
         treatment_type = as_factor(treatment_type),
         availability_of_vaccines_treatment = as_factor(availability_of_vaccines_treatment)
  )

gaq <-
  gaq |>
  mutate(country = as_factor(country),
         date = as.Date(date)
  )
```

```

#deleting other disease_category than respiratory for ghs
ghs <- ghs %>%
  filter(disease_category == "Respiratory")

#Only rows from 2014 or later for ghs
ghs <- ghs %>%
  filter(year >= 2014)

#deleting rows with countries that are not in both datasets.
ghs <- ghs %>%
  filter(!country %in% c("Italy", "Canada", "Argentina"))

gaq <- gaq %>%
  filter(!country %in% c("Bangladesh", "Egypt", "Ethiopia", "Pakistan", "Phillipines", "Thailand", "Vietnam"))

#Deleting states and cities, by taking the average from them to only have countries.
gaq_country_year <- gaq %>%
  mutate(year = as.numeric(format(date, "%Y"))) %>%
  group_by(country, year) %>%
  summarise(
    across(where(is.numeric), ~ mean(.x, na.rm = TRUE)),
    .groups = "drop"
  )

#Deleting useless columns for gaq
gaq_country_year <- gaq_country_year %>%
  select(-any_of(c("wind_speed_km_h",
    "humidity_percent",
    "deforestation_rate_percent",
    "aqi")))

#Making year the correct datatype for both datasets.
gaq_country_year <- gaq_country_year %>%
  mutate(year = as.numeric(year))

ghs <- ghs %>%
  mutate(year = as.numeric(year))

#Making the names of countries the same in both datasets
ghs$country <- countrycode::countrycode(
  ghs$country,
  origin = "country.name",
  destination = "country.name"
)

gaq_country_year$country <- countrycode::countrycode(
  gaq_country_year$country,
  origin = "country.name",
  destination = "country.name"
)

#Making the first variable. In generate necessary date we explain all variables.

```

```

gaq_country_year <- gaq_country_year %>%
  mutate(
    total_air_pollution = rowMeans(
      scale(select(.,
        pm2_5_ug_m3, pm10_ug_m3,
        no_ug_m3, no2_ug_m3,
        n_ox_ppb, nh3_ug_m3,
        co_mg_m3, so2_ug_m3,
        o3_ug_m3, benzene_ug_m3,
        toluene_ug_m3, xylene_ug_m3
      )),
    na.rm = TRUE
  )
)

#Selecting only the variables that we want to merge for the merged dataset
gaq_clean <- gaq_country_year %>%
  select(country, year, total_air_pollution)

#Merging the datasets
result <- ghs %>%
  inner_join(gaq_clean, by = c("country", "year"))

#Making variables 2 and 3. In generate necesarry date we explain all variables
result <- result %>%
  mutate(
    disease_intensity = as.numeric(scale(population_affected)),
    disease_per_pollution = population_affected / (abs(total_air_pollution) + 1)
  )

#saving datasets
saveRDS(gaq_country_year, "gaq_country_year.rds")
saveRDS(ghs, "ghs_country_year.rds")
saveRDS(result, "result.rds")

```

Please use a separate 'R block' of code for each type of cleaning. So, e.g. one for missing values, a new one for removing unnecessary variables etc.

3.2 Generate necessary variables

Variable 1

I already create the variable in data cleaning, because I use functions for data cleaning that contain that variable. So here I explain what they do; in data cleaning, the code is indicated with # to show which code it is.

Variable 1: Total air pollution is an index created by standardising the different pollutants and then summing them. Ultimately, an average is obtained that is compared to the pollution of all countries in the merged dataset.

Value >0: more polluted than average, value = 0: moderately polluted, value 0<: less polluted on average.

Variable 2 en 3 I already create the variable in data cleaning, because I use functions for data cleaning that contain that variable. So here I explain what they do; in data cleaning, the code is indicated with # to show which code it is.

Variable 2: Disease intensity shows whether people are affected by respiratory diseases more or less than average this year: 1 = much higher disease burden, 0 = average, -1 = much lower disease burden. It takes the average of all countries and rows.

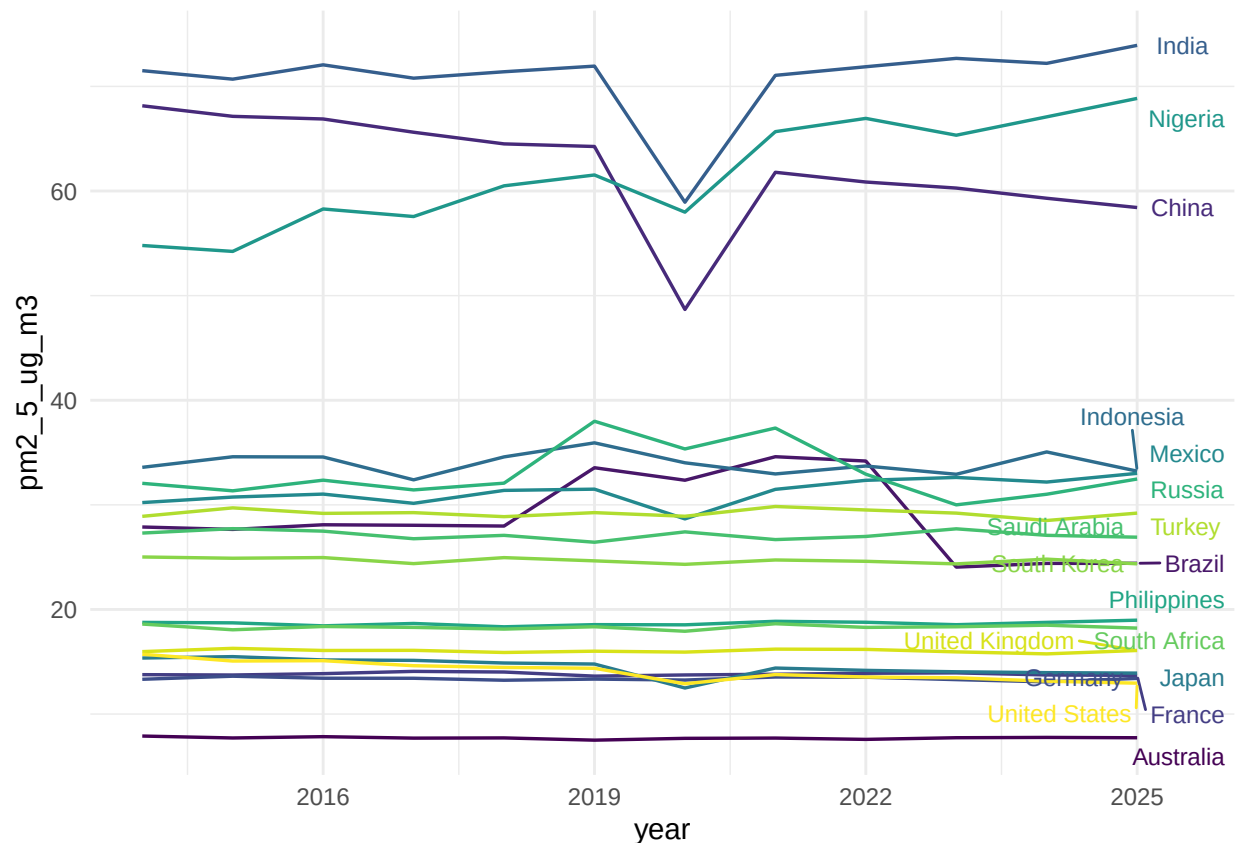
Variable 3: Disease per population ratio, which shows how pollution compares to the number of diseases. Low value = fewer disease cases relative to pollution; high value = a lot of disease cases despite low/medium pollution.

3.3 Visualize temporal variation

```
# data opschoenen
gaq_country_year_clean <- gaq_country_year %>%
  filter(!is.na(pm2_5_ug_m3), !is.na(year))

# last points per country
last_points <- gaq_country_year_clean %>%
  group_by(country) %>%
  filter(year == max(year)) %>%
  ungroup()

# plot
ggplot(gaq_country_year_clean,
  aes(x = year,
    y = pm2_5_ug_m3,
    color = country,
    group = country)) +
  geom_line(linewidth = 0.6) +
  geom_text_repel(data = last_points,
    aes(label = country),
    size = 3,
    nudge_x = 0.5,
    show.legend = FALSE) +
  scale_color_viridis_d() +
  theme_minimal() +
  theme(legend.position = "none")
```



This plot is very useful to see who has the higher pollution. With that we can compare with 3.4 and 3.5 if there is a relation. It also help to understand where the outliers are. We can see that India, China and Nigeria have by far the most pollution in the discussion make the link with the respiratory disease.

3.4 Visualize spatial variation

```
# Wereldkaart ophalen
world <- ne_countries(scale = "medium", returnclass = "sf")

# Gemiddelde score per land berekenen
result_country <- result %>%
  group_by(country) %>%
  summarise(
    disease_per_pollution = mean(disease_per_pollution, na.rm = TRUE)
  )

# Scores opdelen in 5 categorieën
result_country <- result_country %>%
  mutate(
    score = ntile(disease_per_pollution, 5)
  )

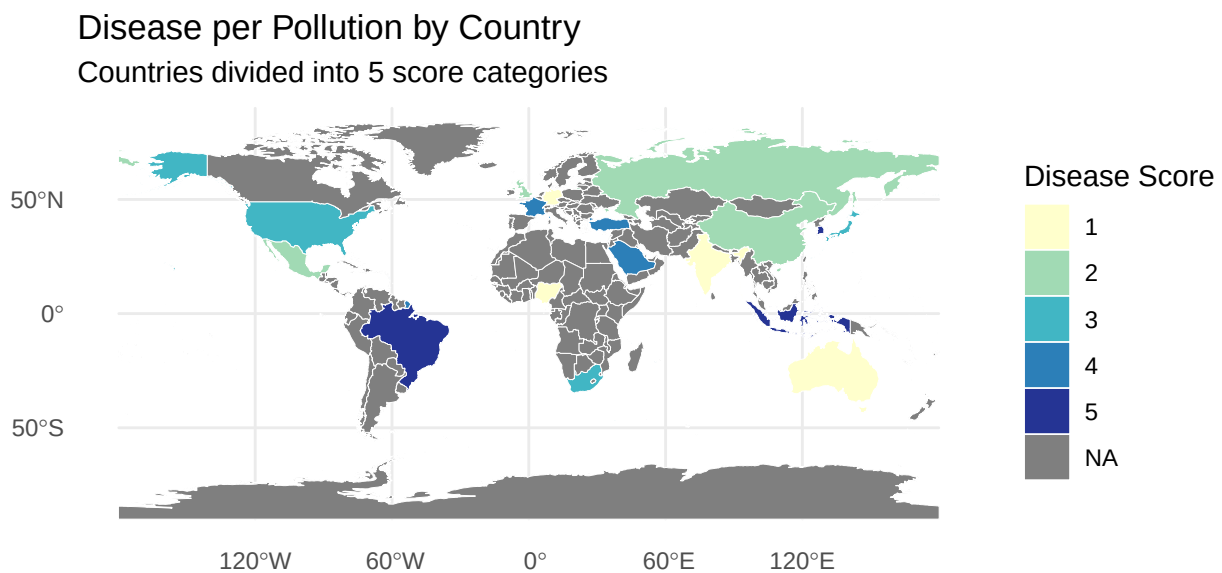
#juiste namen geven
result_country$country[result_country$country == "United States"] <- "United States of America"
```

```

# Landnamen koppelen aan wereldkaart
world_data <- world %>%
  left_join(result_country, by = c("name" = "country"))

# Wereldkaart plotten
ggplot(world_data) +
  geom_sf(aes(fill = factor(score)), color = "white", size = 0.1) +
  scale_fill_manual(
    values = c(
      "#ffffcc", # score 1
      "#a1dab4", # score 2
      "#41b6c4", # score 3
      "#2c7fb8", # score 4
      "#253494"  # score 5
    ),
    name = "Disease Score"
  ) +
  labs(
    title = "Disease per Pollution by Country",
    subtitle = "Countries divided into 5 score categories"
  ) +
  theme_minimal()

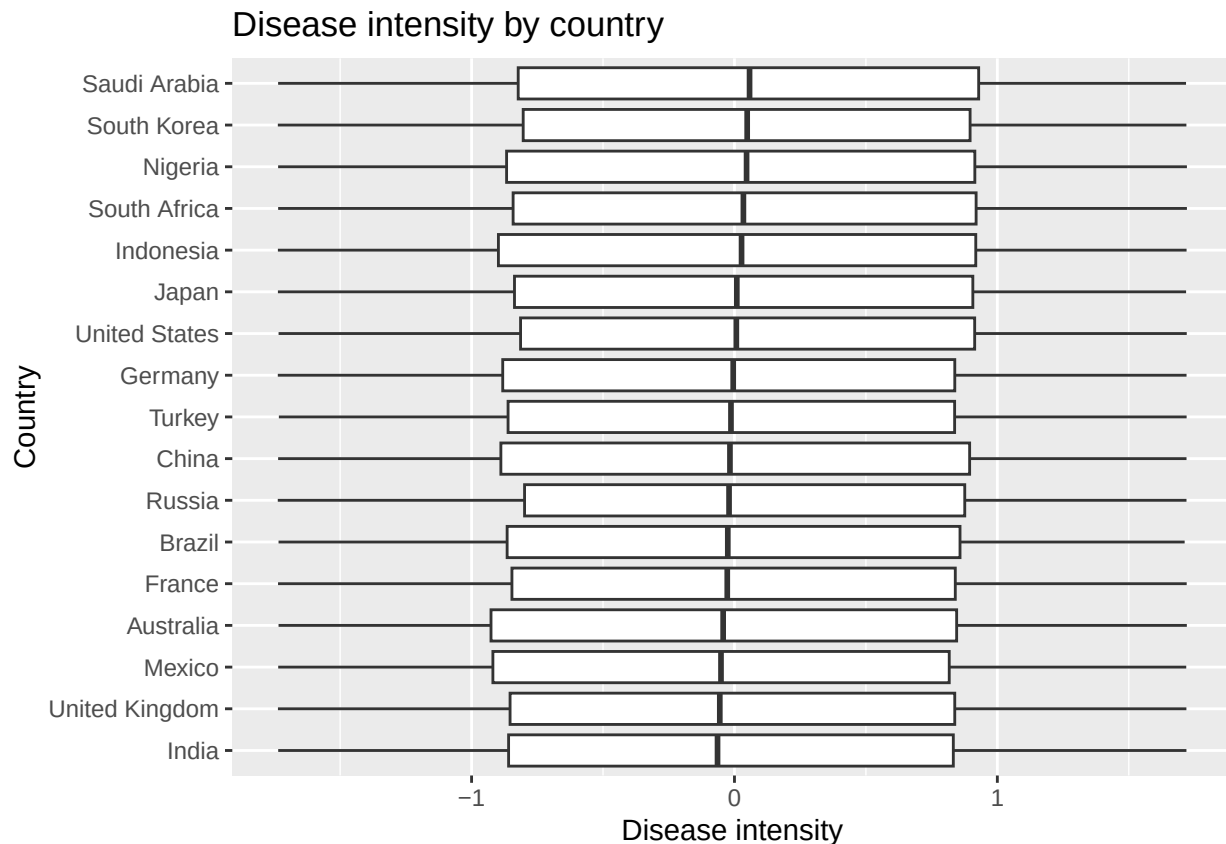
```



This map is very useful to see when higher pollution does not mean a higher respiratory disease. It helps us understand where the outliers are of the relation and makes us think why that is. Brazil has a high score and India and Nigeria a low score. The score is made with the disease per Pollution variable.

3.5 Visualize sub-population variation

```
#box plot
ggplot(result, aes(x = reorder(country, disease_intensity, median),
                    y = disease_intensity)) +
  geom_boxplot() +
  coord_flip() +
  labs(
    title = "Disease intensity by country",
    x = "Country",
    y = "Disease intensity"
  )
```

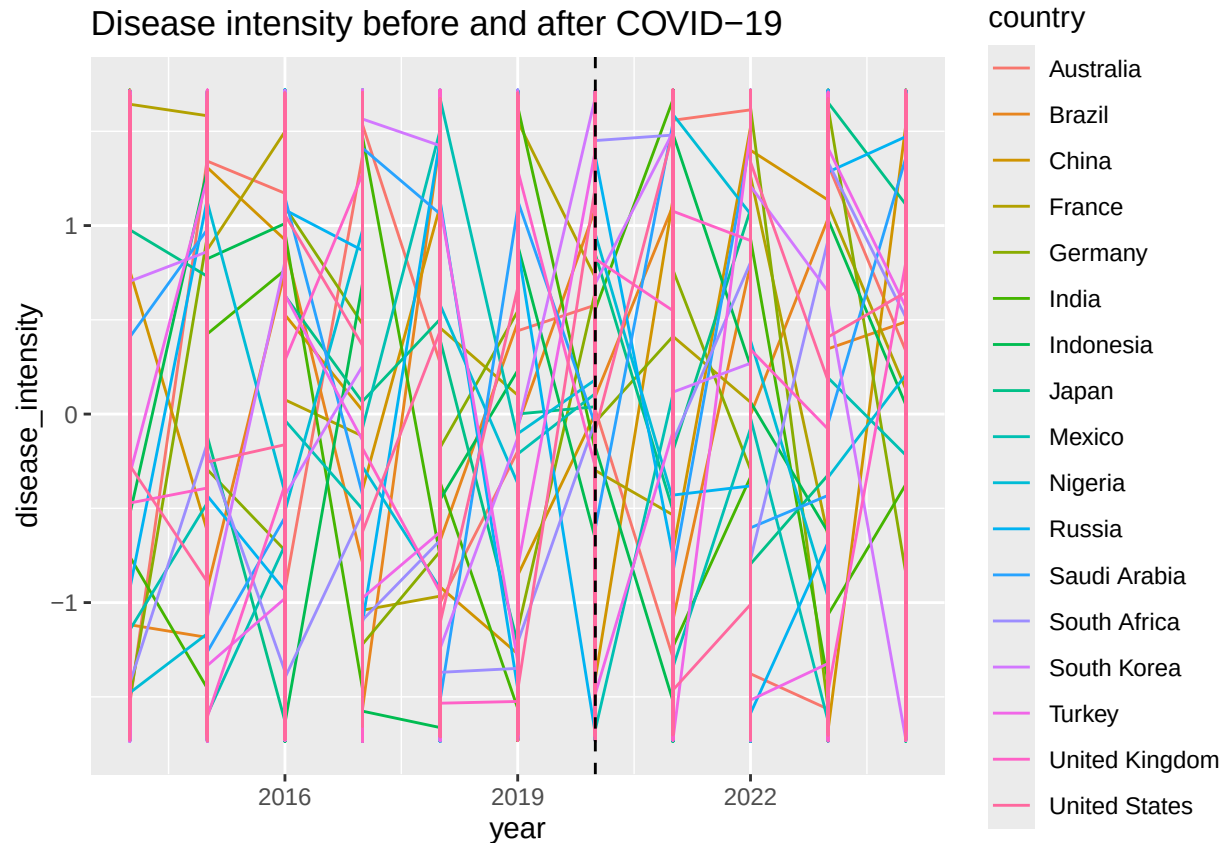


The boxplot reveals a high difference in disease intensity across countries. Russia and South Africa have a high median (the black lines in the boxes) compared with other countries; this suggests a higher disease burden than in other countries. Variability differs across countries, with some having a more unstable disease intensity (as seen by the size of the box). There are some outliers suggesting extreme spikes in disease intensity (indicated by the dots), which may be temporary shocks or faults in the measurements. The results show a high variability between countries.

3.6 Event analysis

Analyze the relationship between two variables.

```
# event plot
ggplot(result,
  aes(year,
    disease_intensity,
    color = country)) +
  geom_line() +
  geom_vline(xintercept = 2020,
    linetype = "dashed") +
  labs(title = "Disease intensity before and after COVID-19")
```



In this graph, you can see an interesting effect after the covid pandemic. You can see that in the COVID pandemic and afterwards, the disease intensity is lower than before. This may seem odd, after all it is a respiratory disease, but this can be because a lot of people stopped working and stayed at home, so they would not be exposed to a lot of pollution. Another reason can be that there were fewer people working, so there could be less output. You can see in the graph from 3.3 that there was a big drop in pollution in some countries, notably India and China. So that is interesting to see. We cannot draw further conclusions per country because it is hard to see which line is from which country.

Part 4 - Discussion

4.1 Discuss your findings

Our findings are interesting to say the least. When you see in the time graph(3.3) India, China and Nigeria are the most polluted. When you look at their disease intensity(plot 3.5) , India and China are low, and

Nigeria is high on the list of disease intensity. This would not suggest a relation between air pollution and respiratory diseases, but you should realise that India and China have more than 1 billion people as residents. It could be that not every case of respiratory disease is discovered because of the huge size of the population. Russia, Mexico and Indonesia are next on the air pollution list, which Russia is the highest disease intensity, and Indonesia is 6th. These are averages as Nigeria has the highest in variability(farthest line in 3.5), and India also has a high variability in higher or lower disease intensity, while Mexico has a really high on the variability line. From this, you can conclude that higher air pollution most of the time means more respiratory disease cases. You can see in the of 3.4 when that is not the case Spending more money on respiratory diseases can work, as we see when COVID came, there were higher budgets for healthcare, which resulted in lower respiratory disease intensity. Spending less time outside also helps reduce the intensity of lower respiratory disease.

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: <https://github.com/jespersmit/programming-groep>

5.2 Reference list

European Environment Agency. (2024, January 25). Air pollution. [Www.eea.europa.eu. https://www.eea.europa.eu/en/topics/in-depth/air-pollution](https://www.eea.europa.eu/en/topics/in-depth/air-pollution)

Levine, S., Marciniuk, D., Aglan, A., Celedón, J., Fong, K., Horsburgh, R., Malhotra, A., Masekela, R., Mortimer, K., Redde, H., Rice, M., & Simonds, A. (n.d.). UT Health-San Antonio and South Texas Veterans Health Care System. https://firsnet.org/wp-content/uploads/2025/01/FIRS_Master_09202021.pdf

Malaiaarasugraj. (z.d.). Global health statistics [Dataset]. Kaggle. <https://www.kaggle.com/datasets/malaiaarasugraj/global-health-statistics>

ashyou09. (z.d.). World air pollution and AQI dataset (2014–2025) [Dataset]. Kaggle. <https://www.kaggle.com/datasets/ashyou09/world-air-pollution-and-aqi-dataset-2014-2025/data>